Demand forecasting and supply optimisation

The case of South-Africa

Presented by Bérangère & Siyao

Outline

1. Context
2. Problem Definition
3. Literature Review
4. Data & Methodology
5. Demand Forecasting
6. Supply Optimization
7. Limitations
8. Next steps

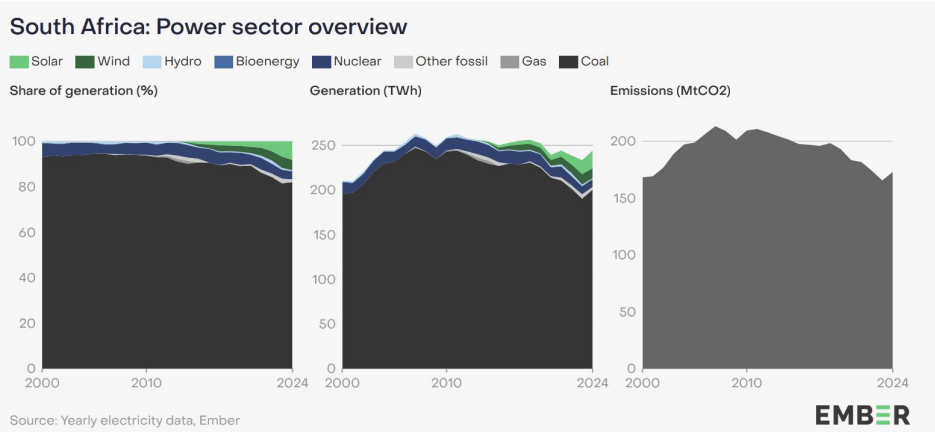
Context - South Africa Power System Under Structural Stress

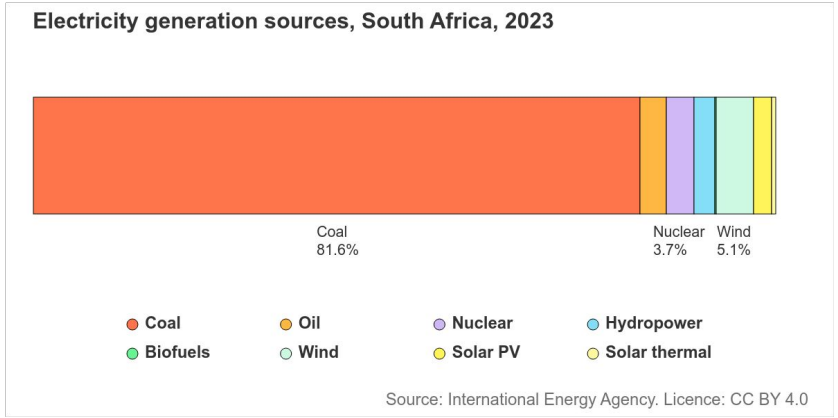
Why is south africa a special case?

Electricity shortages in South Africa are structural, not exceptional

- **Supply side**: Coal-dominated (~80%) system with aging fleet → availability constraint
- **Demand side**: Observed demand constrained by load shedding → downward bias in demand data
- **Flexibility**: Rising renewables, limited system flexibility → dispatch rigidity

Load shedding reflects equilibrium under capacity scarcity





Context - System dynamics shaping demand

1. **Observed demand is evolving and constrained**

- Energy demand evolves rapidly and does not fully reflect underlying needs
- Recent demand decline (2023-2024, CSIR) driven by the rapid expansion of rooftop PV and decrease of GDP growth rate since 2021.
- Suppressed demand under load shedding biases historical consumption data

2. **Structural and infrastructure constraints**

- Aging, coal-fired power plants with declining availability
- Chronic under-maintenance and long-term under-investment
- Limited system flexibility constrains real-time dispatch

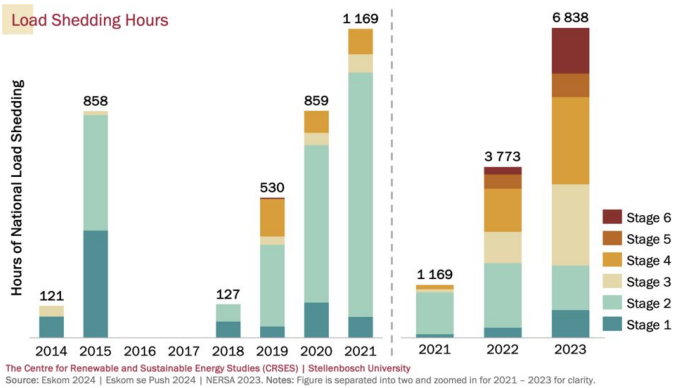
3. **Policy transition as a background, not a short-run solution**

- Just Energy Transition Partnership (JETP)
  Limited short-run impact on capacity constraints

Problem Definition - Objective

South Africa's Electricity crisis:

- Power outages
- Load shedding



How can we reconcile electricity demand with available supply under technical, economic, and contractual constraints?

Approach

1) Demand forecasting

- Estimate the true hourly electricity demand

- Capture seasonal trends

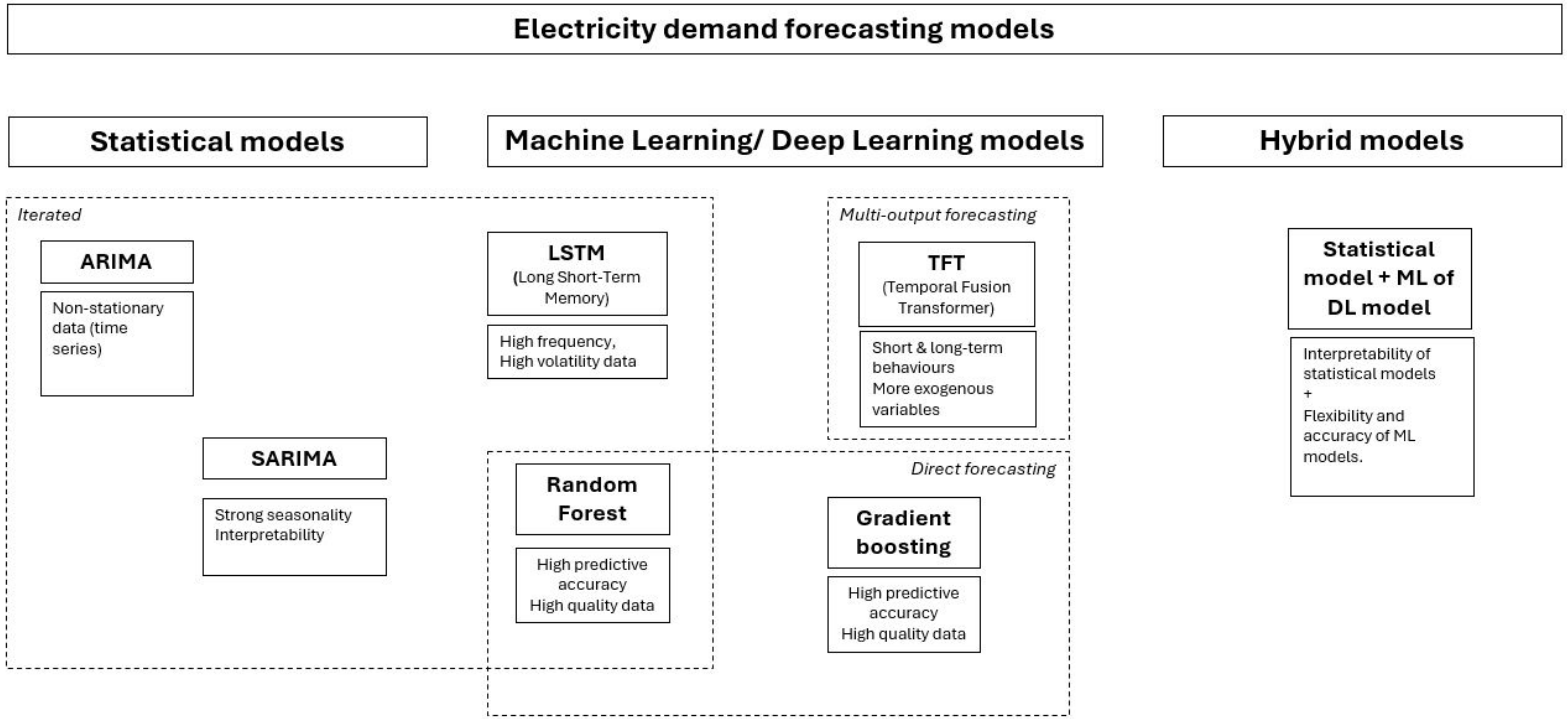
- Provide a reliable baseline for system planning

2) Supply optimisation

- Determine how generation capacity should be dispatched

- Identify marginal technologies and resulting electricity price

Literature review



Classification of energy demand forecasting methodologies
Computed by the authors

No single forecasting approach dominates across all settings: model performance depends on **data quality, system constraints, and the operational objective**.

Methodology and data

Study period:

April 1, 2021 to March 31, 2025

Key variables:

- **Demand and forecast variables:** Hourly electricity consumption and modeled demand
- **Supply by technology:** Coal, gas, nuclear, hydro, solar, wind
- **Renewables and installed capacity**
- **Outages & availability:** Forced outages, load shedding

Modeling approaches

Statistical Models - ARIMA, SARIMA
Machine Learning Models - LSTM, TFT
Hybrid Model - SARIMAX + LSTM

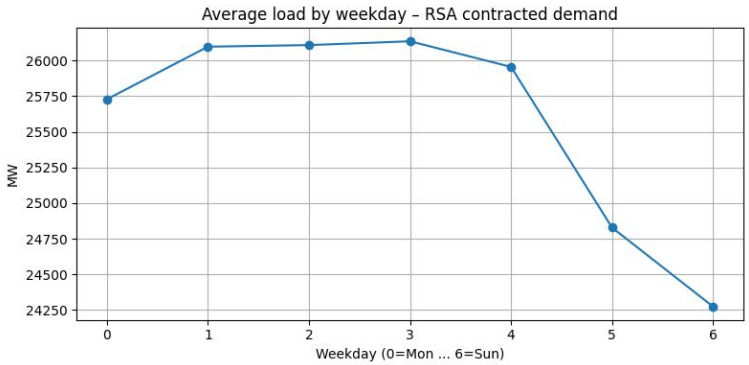
Methodology and data

Data cleaning and normalization:

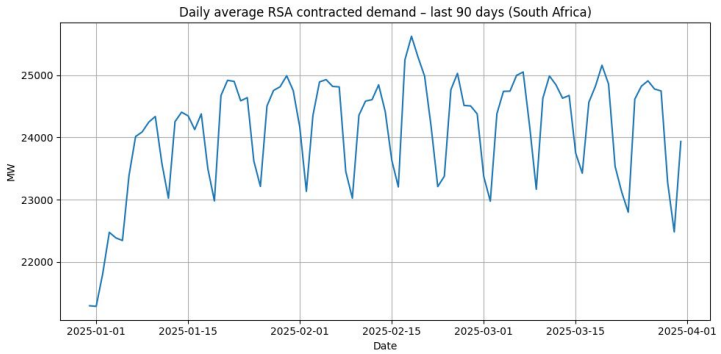
- Convert into proper time-date format
- Set the **time index**

Key variables: RSA contracted demand
Residual demand

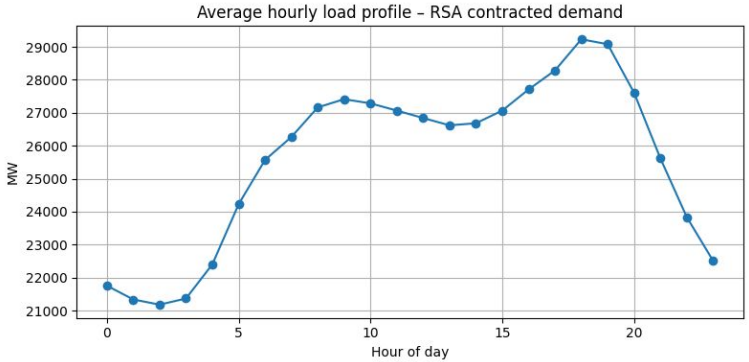
Load curves: Show system demand per hour, week, year



Higher demand on weekdays: More industrial and commercial activity, but small gap → Mining



Short-term shocks (e.g. heat waves, public holidays).



Typical daily shape: Night valley, Morning ramp, Evening peak.

Methodology and data

Weekly load curve: **Shows how daily life in South Africa shapes electricity demand**

Why do we observe peaks?

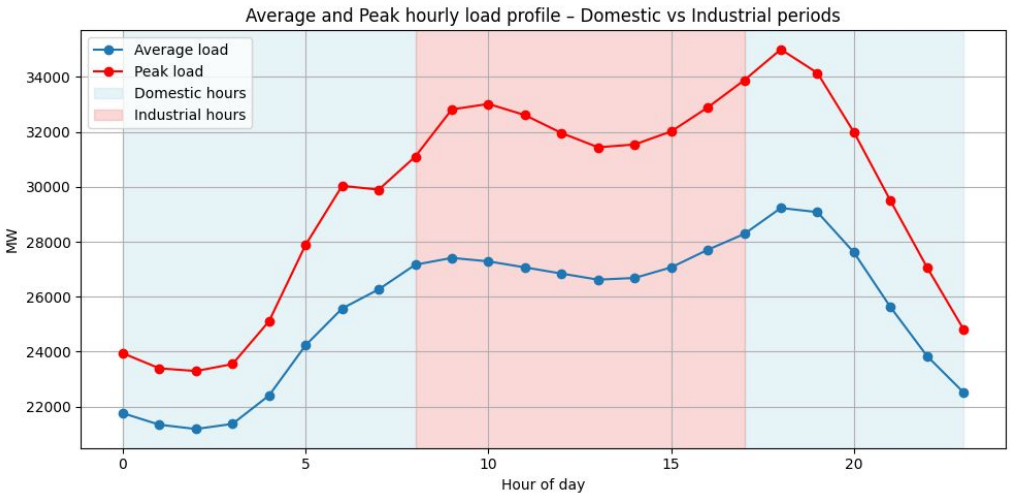
Daytime peak (8am-5pm): Driven by industrial, commercial, and office activity

Domestic peaks:

- Morning (5am-6am): Cooking, appliance
- Evening (6pm): Cooking, TV, appliances

What is the impact of peak hours?

Electricity is **most expensive during peak periods** → Implications for consumers and system planning.

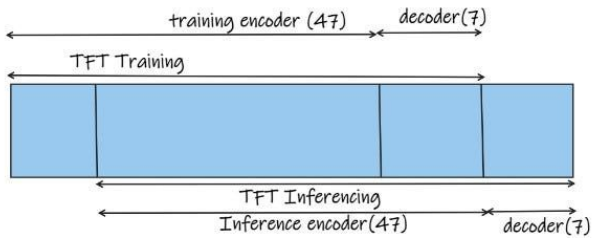


Methodology and data

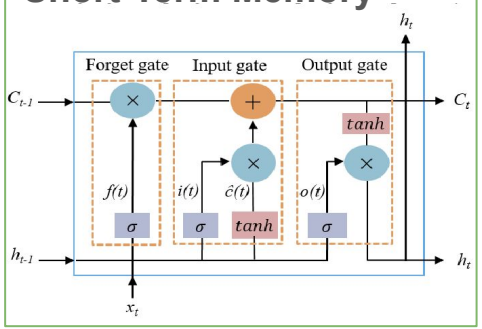
Statistical models (Seasonal) AutoRegressive Integrated Moving Average

1. **ARIMA:** Predicts future demand using past values, past changes, and past errors.
2. **SARIMA:** ARIMA + Seasonality

3. TFT (Temporal Fusion Transformer)

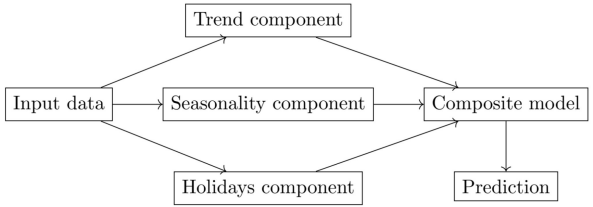


4. LSTM - Long Short-Term Memory



5. Prophet

By Meta for business forecasting.



6. Hybrid - SARIMAX + LSTM

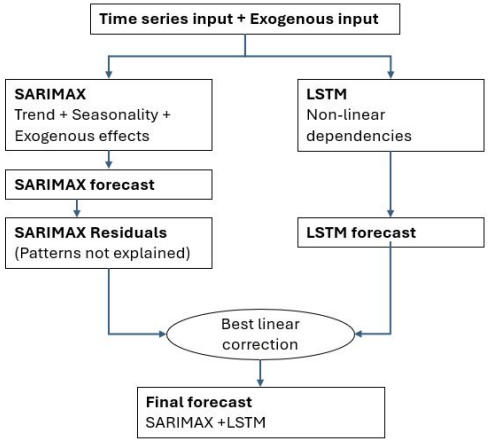


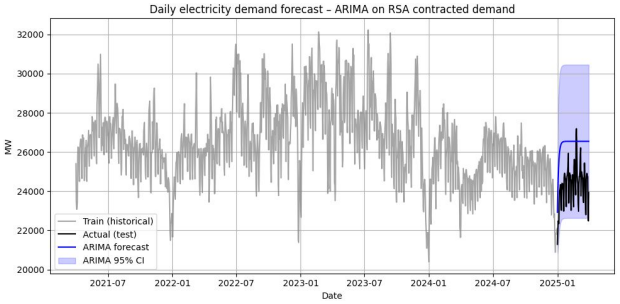
Diagram of the hybrid model
Computed by the authors

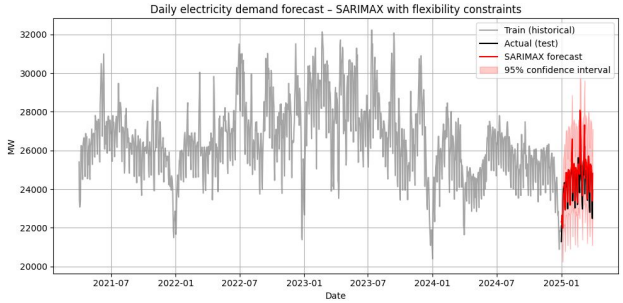
Demand forecasting - Task Definition and Evaluation Setup

Forecasting must account for suppressed demand and asymmetric risks

1. **What we forecast (and why it's hard)**
   - Short-run demand (daily; test = last 90 days)
   - Strong weekly seasonality + high volatility
   - Observed demand is affected by load shedding → suppressed signals
2. **What we use as drivers (system-available)**
   - Load shedding: MLR / ILS (intensity proxy)
   - Availability shocks: PCLF/UCLF/OCLF (planned/unplanned/outage factors)
   - Flexibility proxy: pumped generation hours
   - Calendar effect: weekend indicator
3. **How we evaluate**
   - Point forecasts + 95% prediction intervals (ARIMA/SARIMAX)
   - **Baseline accuracy:** MAE / RMSE / MAPE
   - **Risk-aware evaluation:** tail errors and asymmetric loss (pinball)

ARIMA/SARIMAX: 95% confidence interval





LSTM: Train-test split: Test: Last 90 days

Demand forecasting - Model Roles

Model strengths

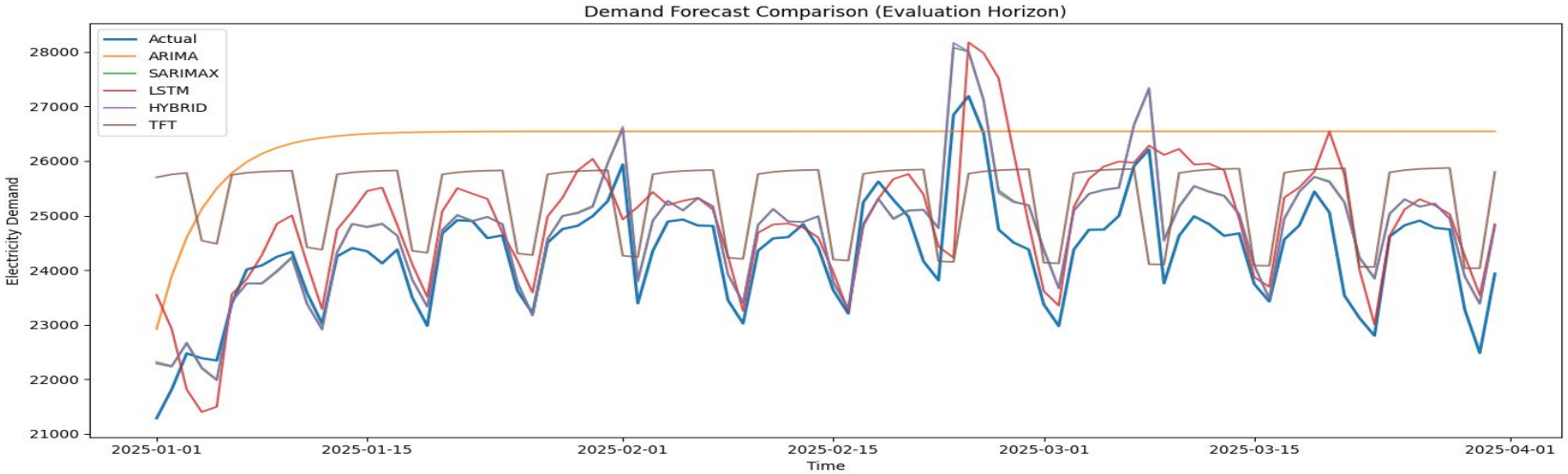
- SARIMAX: captures stable weekly seasonality and operational constraints
- LSTM: captures nonlinear deviations and extreme trajectories
- TFT: captures complex temporal interactions but requires large, stable datasets

Roles in the Final Framework

- SARIMAX → baseline demand scenario
- LSTM → stress-test and uncertainty exploration
- TFT → robustness and complexity benchmark

Hybrid approach

- Central path: SARIMAX forecast
- Deviation band: LSTM-based correction capturing nonlinear stress
- Usage: robustness check rather than point prediction



Demand forecasting - Key Findings

The evaluation logic

- **MAE / RMSE**: baseline accuracy (average vs large deviations)
- **Pinball loss (q = 0.95)**: penalizes under-forecasting (blackout risk)
- **Tail metrics (P95 / max error)**: severity of extreme failures

Ranking logic

- Models are ranked by tail-risk metrics (Pinball loss q=0.95, 95th percentile absolute error), not by MAE alone.

→ Models are compared by their contribution to **risk management**, not **point accuracy** alone.

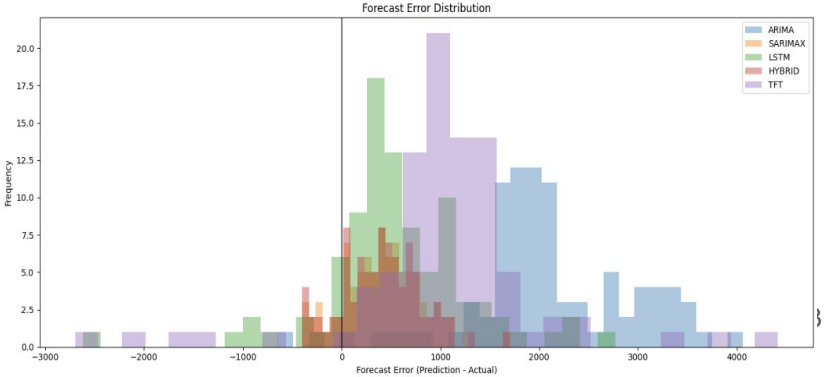
Conclusions

- In the short run, simpler econometric models outperform complex ML approaches
- SARIMAX remains robust not only on average, but also in the upper tail
- Hybrid correction improves accuracy only marginally
- LSTM and TFT perform significantly worse under capacity stress

Takeaways

- Forecast quality depends more on data structure and tail behavior
- This result directly informs how demand should be modeled in the supply optimization stage

	Model	MAE	AE_P95	AE_Max	Pinball_q95	Bias
1	SARIMAX	473.115656	1035.324727	1713.679817	52.310671	409.438127
3	HYBRID	471.203055	1040.725512	1702.189039	52.998070	405.785461
2	LSTM	725.322230	2009.849889	2772.809565	107.258312	567.561784
0	ARIMA	2111.945173	3476.237253	4061.715035	115.065271	2090.905146
4	TFT	1239.046296	2346.917921	4415.302047	148.362800	1047.022995
5	PROPHET	1108.406787	2155.075546	2528.701000	160.032327	875.935702



Supply model

1. Factors influencing dispatch decision

Generation costs - Fuel, O&M

Demand variability - Weather, economic activity, peak hours)

Renewable integration - High intermittency

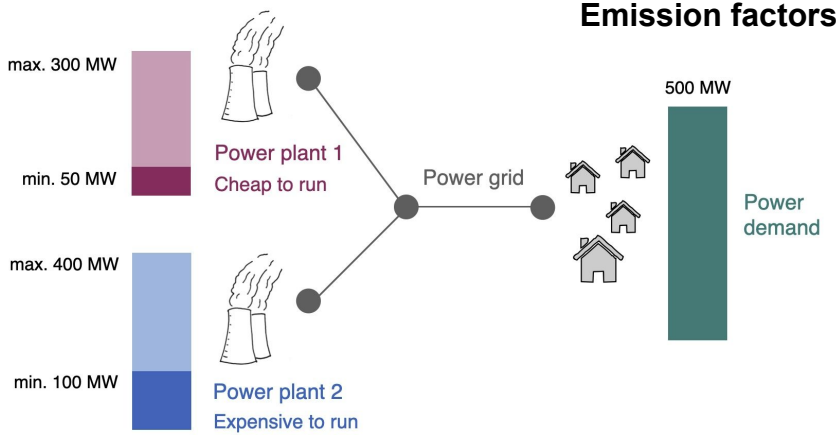
2. Core variables in the supply model

Demand - Hourly electricity requirement to be met

Installed capacity - Maximum available output for each technology

Marginal cost - Cost of producing one additional MWh, key determinant of dispatch order

Emission factors - Technology-specific CO2 intensity



Source: https://www.modelling-energy-systems.org/stable/basics/dispatch-lp.html

Supply model

Objective function: Minimize total system cost, composed of:

- **Generation costs:** Fuel + Variable O&M for each technology
- **Carbon costs** : Emissions and exogenous carbon price
- **Load-shedding penalties**
- **Take-or-pay shortfall penalties**

Costs → [Objective Function] ← Constraints
↓
Optimal Dispatch

Model constraints

- Demand balance

$$\sum_i P_{i,t} \geq D_t$$

- Capacity limits & Renewable availability

$$0 \leq P_{i,t} \leq \overline{P}_i \qquad P_{solar,t} \leq \overline{P}^{avail}_{solar,t}$$

- Take or pay contract constraint

$$Pcontract \leq Pi,t$$

```
shed_cost = 5000     # VOLL/MWh #Nova
Economics for NERSA, 2020
Qmin_MW = 1_500      # MW minimum delivery
obligation
ToP_penalty = 9530   # R/MWh shortfall
penalty
```

Results supply optimization - Economic dispatch

Prices computation:

Economic dispatch model (Pyomo)
No load shedding allowed

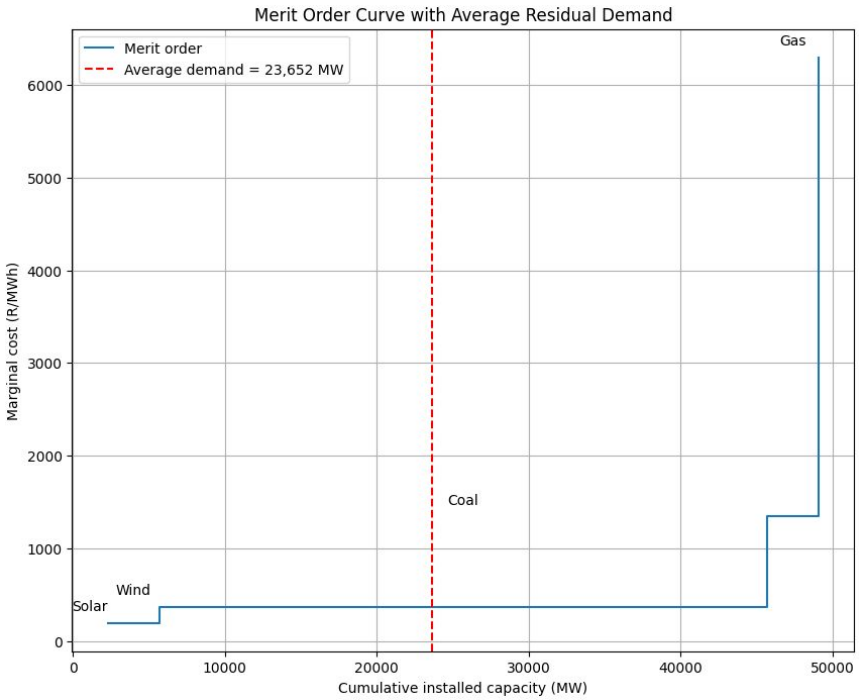
System Marginal Price (Marginal cost of the marginal technology at every hour): 601.11 R/MWh.

Merit order curve: Static representation of supply ranked by marginal cost. Determines which power plant will produce and deliver electricity to meet demand.

Renewables: Called first, low-cost

Coal: Dominant source, price-setting technology

Gas: Always dispatched at full capacity (**3,400 MW**)



Limitations

Demand side:

- Focus on a **limited set of system-available drivers** to ensure operational realism
- Observed demand reflects **constrained consumption** rather than latent demand under unconstrained supply
- **Weather and price** feedback effects are not explicitly modeled at this stage
- **Richer demand drivers** could improve precision in an extended framework

Supply side:

- Focus on **short-run dispatch** decisions under data constraints
- **Nuclear and hydro** generation not yet integrated in the current version
- **Operational constraints** are simplified at this stage, including: ramping limits, coal must-run requirements, and renewable curtailment

These limitations reflect scope choices to prioritize **transparency, robustness, and policy relevance** in the short run.

Next steps

1. **Pricing analysis:**

➔ Baseline
➔ +5% renewables penetration
➔ +25% renewables penetration

2. **Model extension : Nuclear Integration**

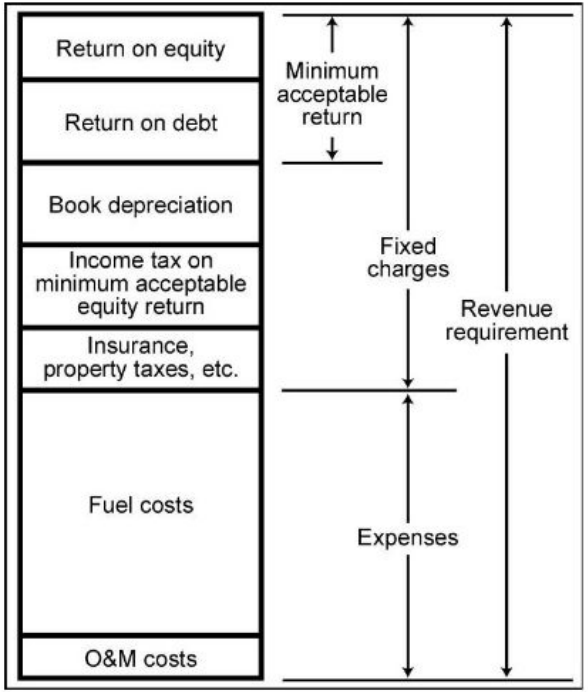
Add nuclear power as an **exogenous must-run baseload**

3. **Improving the forecasting framework**

Incorporate additional exogenous variables for higher accuracy (Temperature, Economic activity)
To better capture true underlying demand and further reduce forecasting error

4. **Policy implications**

Provide actionable insights for planners:
➔ Where to add capacity
➔ Which technologies to expand or refurbish
➔ How to reduce reliance on high-cost marginal plants



Annex - Problem definition

Stages of loadshedding and their impact on the end user[35][36]

Stage	Energy load removed from the national grid	Typical Impact	Percentage of grid users without power	An example loadshedding area in Cape Town might experience power cuts at the following times
Stage 1	<1000 MW	6h over 4 days (6/96h)	~6% without power	16:00–18:30
Stage 2	<2000 MW	12h over 4 days (12/96h)	~12.5% without power	00:00–02:30 & 16:00–18:30
Stage 3	<3000 MW	18h over 4 days (18/96h)	~19% without power	00:00–02:30 & 16:00–18:30
Stage 4	<4000 MW	24h over 4 days (24/96h)	~25% without power	00:00–02:30 & 08:00–10:30 & 16:00–18:30
Stage 5	<5000 MW	30h over 4 days (30/96h)	~31% without power	00:00–02:30 & 08:00–10:30 & 14:00–18:30
Stage 6	<6000 MW	36h over 4 days (36/96h)	~37% without power	00:00–02:30 & 08:00–10:30 & 14:00–18:30
Stage 7	<7000 MW	42h over 4 days (42/96h)	~44% without power	00:00–02:30 & 08:00–10:30 & 14:00–18:30 & 22:00–00:30
Stage 8	<8000 MW	48h over 4 days (48/96h)	~50% without power	

Source: https://loadshedding.eskom.co.za/LoadShedding/ScheduleInterpretation

Annex – Key References

A. Demand Forecasting (Reviews & Econometric Baselines)

- Ugbehe, U. O., Waziri, M. Y., Abba, O. M., & Elameyi, P. J. (2025). Electricity demand forecasting methodologies and applications: a review. Sustainable Energy Research, 12, 19.
- Adeoye, O., & Spataru, C. (2019). Modelling and forecasting hourly electricity demand in West African countries. Applied Energy, 242, 311-333.

B. Deep Learning & Hybrid Forecasting

- Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. International Journal of Forecasting, 37(4), 1748-1764.
- Mazibuko, T., & Akindeji, K. T. (2025). Hybrid Forecasting for Energy Consumption in South Africa: LSTM and XGBoost Approach. Energies, 18(16), 4285.
- Athanasopoulou, M. E., et al. (2022). A Hybrid Model for Forecasting Short-Term Electricity Demand. Proceedings of the 3rd ACM International Conference on AI in Finance (ICAIF '22).

C. System Context & Policy (Sub-Saharan Africa / South Africa)

- Marong, L. K., et al. (2018). The Gambia's future electricity supply system: Optimizing power supply for sustainable development. Energy Strategy Reviews, 22, 235-249.
- Kalima, M. (2025). South Africa's pledge to decommission coal plant flounders. The Energy Pioneer.

Annex - Methodology

Statistical models (Seasonal) AutoRegressive Integrated Moving Average

1. **ARIMA (p,d,q)**

$$\Delta^d y_t = c + \sum_{i=1}^{p} \varphi_i \Delta^d y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

2. **SARIMA (p,d,q) (P,D,Q)**

$$\Delta^d \Delta_s^D y_t = c + \sum_{i=1}^{p} \varphi_i \Delta^d \Delta_s^D y_{t-i} + \sum_{k=1}^{P} \Phi_k \Delta^d \Delta_s^D y_{t-ks} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \sum_{m=1}^{Q} \Theta_m \varepsilon_{t-ms}$$

Annex - Supply model

```
cost = {"coal":83, "gas":388, "solar":12, "wind":22}        # Marginal cost in R/MWh #EPRI
emissions = {"coal":0.9, "gas":0.4, "solar":0.0, "wind":0.0}         # tCO2/MWh #standard
carbon_price = 0    # Because of the Phase I of carbon tax
shed_cost = 5000    # VOLL/MWh #Nova Economics for NERSA, 2020
lambda_loss = 0.10      # transmission loss #World Bank 2023
allow_shedding = False

Qmin_MW = 1_500     # MW minimum delivery obligation
ToP_penalty = 9530  # R/MWh shortfall penalty
```